METRIC

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MILITARY SPECIFICATION

PUMP, SUBMERSIBLE, BILGE, AXIAL FLOW; ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric-motor-driven, axial flow (mixed flow), submersible, bilge pump (for water), herein after referred to by the term "pump". The pump operates in both horizontal and vertical positions and is designed for use in military vehicles equipped with nominal 24 volt (V) direct current (dc) electrical systems. At 28 V dc, the pump shall deliver not less than 174 liters per minute (L/m) at 1.8 meter (m) static pressure head (see 6.1, 6.4.1, and 6.5).

1.2 Configuration. The pump covered by this specification is available in five configurations. The part numbers and corresponding descriptions are listed below (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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Part number	Description
8763300	With inlet elbow, without inlet screen.
8763300-1	Without inlet elbow, without inlet screen, with discharge elbow.
8763300-2	Without inlet elbow, without inlet screen, without discharge elbow.
8763300-3	Same as 8763300-2 except with a 90 degrees (°) electrical connector.
8763300-4	Same as 8763300-1 except with a 90° electrical connector.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-F-13927 - Fungus Resistance Test; Automotive Components.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

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MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-45662	- Calibration Systems Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8763300	- Pump Assembly Bilge.

(Copies of the drawing listed above required by the offerors/contractor in connection with specific acquisition function should be obtained from Commander, US Army Tank-Automotive Command, ATTN: Contracting Officer, Warren, MI 48397-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish samples, of the specified part number (see 6.2), which shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. Materials used in the fabrication of the pump shall be as specified herein and in accordance with referenced standards, specifications and drawing (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.2).

3.3 Design and construction. The pump shall be designed and constructed as specified herein and in accordance with Drawing 8763300 for the applicable part number (see 6.2). The pump shall operate in either a vertical (motor above the pump) or horizontal (with a 90° elbow) position. The pump shall be of such configuration as to permit the centerline of the propeller shaft to be within 6.3 cm of the mounting surface when installed in the horizontal position (see 4.8.1 and 4.8.2).

3.3.1 Weight. The weight of the pump shall be not greater than 5.4 kilograms (kg) (see 4.8.2).

3.3.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. The use of dissimilar metal couples shall be in accordance with MIL-STD-889 (see 4.8.2).

3.3.3 Standard parts. Military standard parts shall be used wherever possible (see 4.8.2).

3.3.4 Threaded surfaces. All screw threads shall conform to FED-STD-H28. The thread form, class and number of threads per inch shall be as specified on the applicable standards or drawing (see 4.8.1 and 4.8.2).

3.3.5 Electric motor. Unless otherwise specified (see 6.2), the motor shall be a direct current, series type and rated at 28 V for operation in an ambient temperature of 21°C. The motor shall have provision for quick replacement of the brushes (see 4.8.2).

3.3.5.1 Brushes. The motor shall be provided with high grade brushes capable of operating continuously for not less than 500 hours (h) without replacement (see 4.8.1).

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3.3.5.2 Power rating. The continuous-duty rating of the motor shall be not less than the brake horsepower requirement of the pump under any condition of head and capacity, from the rated output to free delivery (see 4.8.1).

3.3.5.3 Bearings. The motor shall be provided with anti-friction lubricated-for-life, sealed bearings, conforming to FF-B-171, which shall function on a continuous-duty basis (see 4.8.1).

3.3.5.4 Temperature rise. The temperature rise of the motor, measured on the outside surface (base metal) of the housing over the poles when running in air at 25°C, shall be not greater than 71°C whether its pumping or running dry (see 4.8.2).

3.3.6 Interchangeability. To provide for interchangeability, all assemblies and components having the same part number shall be functionally and dimensionally interchangeable (see 4.8.2).

3.4 Performance.

3.4.1 Water level. The pump shall begin pumping with the water level as low as 2.5 cm above the mounting surface (inlet side) of the pump for vertical (shaft) installations; and for horizontal (shaft) installations, the pump shall begin pumping when the water level reaches the centerline of the propeller shaft (see 4.8.3.1).

3.4.2 Rated capacity. The pump shall deliver not less than 174 L/m against a 1.8 m static pressure head through a 32 millimeter (mm) pipe outlet and the pump shall meet or exceed the requirements shown on the performance graph of Drawing 8763300 (see 4.8.3.2).

3.4.3 Dielectric strength. The pump shall show no evidence of puncture of solid insulation, continuing sparkover, breakdown of the dielectric material, or damage to the insulation sufficient to cause failure of the motor and shall meet the requirements of 3.4.2; after being exposed to a 500 V root-mean-square (rms) and 60 hertz (Hz) potential for 1 minute (see 4.8.3.3).

3.4.4 Environmental.

3.4.4.1 Temperature. The pump shall operate in a temperature range of 0 to plus (+) 52 degrees Celsius (°C) (see 4.8.1).

3.4.4.2 Waterproofness. The pump shall evidence no moisture within the motor after being subjected to the waterproofness test specified in 4.8.3.4.1.

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3.4.4.3 Fungus resistance. The pump shall evidence no microbial growth that would adversely affect performance and shall subsequently meet the requirements of 3.4.2 after being subjected to the fungus test specified in 4.8.3.4.2.

3.4.4.4 Shock resistance. The pump shall evidence no permanently deformed, cracked, or broken parts and shall meet the requirements of 3.4.2 after being subjected to sawtooth waveform shock pulses of 30 gravity units (g) for 11 milliseconds (ms) time duration along each of the three major axes (see 4.8.3.4.3).

3.4.4.5 Vibration resistance. The pump shall evidence no permanently deformed, cracked, or broken parts and shall meet the requirements of 3.4.2, after exposure to simple harmonic motion having an amplitude of 0.8 mm (1.6 mm maximum total excursion) through a frequency range of 10 to 55 Hz in each of the three major axes. The sweep time for the frequency range of 10 to 55 Hz and return to 10 Hz shall be 1 minute (see 4.8.3.4.4).

3.4.4.6 Electromagnetic interference. The pump shall meet the requirements of MIL-STD-461A, Notice 4, for electromagnetic interference requirements (see 4.8.3.4.5).

3.4.5 Endurance. The pump shall retain not less than 90% of the rated capacity specified in 3.4.2 after being operated for 350 h (see 4.8.3.5).

3.5 Finish. Unless otherwise specified (see 6.2), the pump shall be cleaned, treated, primed and painted in accordance with MIL-STD-193. All parts of the pump that are exposed directly to chemical contaminations shall be painted with a coating conforming to MIL-C-46168 or MIL-C-53039 (see 4.8.2).

3.6 Identification marking. Unless otherwise specified (see 6.2), the pump shall be identified in accordance with MIL-STD-130. The marking shall be as specified on Drawing 8763300. Identification plates, if used, shall be photosensitive anodized aluminum and shall conform to MIL-P-514 (see 4.8.2).

3.7 Workmanship. Workmanship shall be of a quality which will assure a product free of cracked, broken, or rusted parts or components; scratched finish, peeled or blistered paint; or any other defect which may affect appearance or performance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

Standard condition	Metric unit	English unit
Temperature	25 ± 10°C	77 ± 18°F
Relative humidity	(uncontrolled room ambient)	
Atmospheric pressure	(site pressure)	

4.4 First article inspection. When first article is required (see 3.1), first article inspection shall be performed on first article preproduction or initial production samples as specified herein.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of six pumps (specimens A through F). Preproduction inspection shall consist of the inspections as specified in table I. Tests shall be conducted in the order specified in table II.

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article		Quality conformance		Control
			Preproduction	Initial production	Examination	Tests	
Materials and construction	3.2, 3.3, 3.3.4, 3.3.5.1, 3.3.5.2, 3.3.5.3, and 3.4.4.1	4.8.1	X	X			X
Defects (see table IV and 4.5.2)	3.3 thru 3.3.5, 3.3.5.4, 3.3.6, and 3.5 thru 3.7	4.8.2	X	X	X		
Water level	3.4.1	4.8.3.1	X	X		X	X
Rated capacity	3.4.2	4.8.3.2	X	X		X	X
Dielectric strength	3.4.3	4.8.3.3	X	X		X	X
Environmental	3.4.4	4.8.3.4	X	X			
Temperature	3.4.4.1	4.8.1	X	X			
Waterproofness	3.4.4.2	4.8.3.4.1	X	X		X	X
Fungus resistance	3.4.4.3	4.8.3.4.2	X				
Shock resistance	3.4.4.4	4.8.3.4.3	X	X			X 1/
Vibration resistance	3.4.4.5	4.8.3.4.4	X	X			X 1/
Electromagnetic interference	3.4.4.6	4.8.3.4.5	X	X			X
Endurance	3.4.5	4.8.3.5	X				

1/ The test is required on one sample per month

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TABLE II. Order of preproduction testing.

Specimen	Title	Requirement	Test
A	Vibration resistance (horizontal) Shock resistance Waterproofness Rated capacity	3.4.4.5 3.4.4.4 3.4.4.2 3.4.2	4.8.3.4.4 4.8.3.4.3 4.8.3.4.1 4.8.3.2
B	Fungus resistance Dielectric strength Water level Rated capacity	3.4.4.3 3.4.3 3.4.1 3.4.2	4.8.3.4.2 4.8.3.3 4.8.3.1 4.8.3.2
C	Vibration resistance (vertical) Shock resistance Electromagnetic interference Rated capacity Endurance (vertical)	3.4.4.5 3.4.4.4 3.4.4.6 3.4.2 3.4.5	4.8.3.4.4 4.8.3.4.3 4.8.3.4.5 4.8.3.2 4.8.3.5
D	Vibration resistance (vertical) Shock resistance Waterproofness Rated capacity	3.4.4.5 3.4.4.4 3.4.4.2 3.4.2	4.8.3.4.4 4.8.3.4.3 4.8.3.4.1 4.8.3.2
E	Fungus resistance Dielectric strength Water level Rated capacity	3.4.4.3 3.4.3 3.4.1 3.4.2	4.8.3.4.2 4.8.3.3 4.8.3.1 4.8.3.2
F	Vibration resistance (horizontal) Shock resistance Waterproofness Rated capacity Endurance (horizontal)	3.4.4.5 3.4.4.4 3.4.4.2 3.4.2 3.4.5	4.8.3.4.4 4.8.3.4.3 4.8.3.4.1 4.8.3.2 4.8.3.5

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select six pumps from the first lot produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I. Tests shall be conducted in the order specified in table III.

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TABLE III. Order of initial production testing.

Title	Requirement	Test
Vibration resistance	3.4.4.5	4.8.3.4.4
Shock resistance	3.4.4.4	4.8.3.4.3
Waterproofness	3.4.4.2	4.8.3.4.1
Dielectric strength	3.4.3	4.8.3.3
Electromagnetic interference	3.4.4.6	4.8.3.4.5
Water level	3.4.1	4.8.3.1
Rated capacity	3.4.2	4.8.3.2

4.4.3 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the pumps from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for tests. Samples for test shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's:

Classification	AQL
Major	1.0
Minor	2.5

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4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance (see 3.3 and 3.3.6).	SIE 1/
102	Threads not as specified (see 3.3.4).	SIE
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Nonconformance in design and construction (see 3.3, 3.3.2, 3.3.3, 3.3.5, and 3.3.5.4).	SIE and Visual
202	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
203	Weight not as specified (see 3.3.1).	SIE
204	Improper finish (see 3.5).	Visual and functional
205	Improper marking (see 3.6).	Visual
206	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table I, and shall conform to an AQL of 4.0 on the basis of % defective. Tests shall be conducted in the following order: 4.8.3.4.1, 4.8.3.1, 4.8.3.3, and 4.8.3.2.

4.6 Control tests. Except as otherwise specified, control tests shall be conducted on 2 pumps from each lot of 100 units consecutively produced, except that not more than 4 pumps shall be tested in a 30-day period. The pump shall be subjected to the control tests specified in table I. Tests shall be performed in the following order: 4.8.3.4.4, 4.8.3.4.3, 4.8.3.4.1, 4.8.3.1, 4.8.3.3, 4.8.3.2, and 4.8.3.4.5.

4.7 Failure. Failure of any pump to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2, 3.3, 3.3.4, 3.3.5.1, 3.3.5.2, 3.3.5.3, and 3.4.4.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3 through 3.3.5, 3.3.5.4, 3.3.6 and 3.5 through 3.7 shall be determined by examination for the defects listed in table IV. Examination shall be visual, functional or by measurement with standard inspection equipment.

4.8.3 Performance.

4.8.3.1 Water level. To determine conformance to 3.4.1, the pump shall be operating and the water level shall be gradually lowered until pumping stops. Without turning the pump motor off, the water level shall be increased until pumping is resumed. The water level at the time pumping begins shall be recorded. After pumping has started, the pump shall demonstrate the ability to continue pumping as the water level is lowered, until a level of 2.5 cm above the mounting surface of the pump inlet is reached.

4.8.3.2 Rated capacity. To determine conformance to 3.4.2, the pump shall be operated while it is (shaft) in the horizontal and vertical position, and the motor at the stabilized operating temperature. The output shall be measured under the following conditions:

a. The static pressure head shall be 1.8 m.
b. The voltage at motor terminals shall be 28 V dc.
c. The ambient air temperature shall be 25 ± 6°C.
d. The specific gravity of water being pumped shall be not greater than 1.03.
e. The water temperature shall be 25 ± 6°C.

4.8.3.3 Dielectric strength. To determine conformance to 3.4.3. the pump shall be tested in accordance with method 301 of MIL-STD-202. The following conditions shall be applicable to this test:

a. Five hundred V rms at a frequency of 60 Hz shall be applied for 1 minute to each appropriate set of points.
b. The test voltage source shall be capable of delivering at least 1 ampere (A) to a 500 ohm resistive load.
c. The potential shall be applied between the armature and ground, and between the field(s) and ground, as applicable. Ground brushes shall be raised from the commutator, and capacitors and any field ground terminal disconnected for the duration of this test.

After the test, the pump shall be tested as specified in 4.8.3.2.

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4.8.3.4 Environmental.

4.8.3.4.1 Waterproofness. To determine conformance to 3.4.4.2, the pump shall be tested in accordance with MIL-STD-1184, type I, class 1, method 100 and procedure 1. Evidence of air leakage at 41.4 kilopascals (kPa) internal air pressure shall not be considered cause for rejection if no moisture is found within the motor at the conclusion of the test.

4.8.3.4.2 Fungus resistance. To determine conformance to 3.4.4.3, the pump shall be tested in accordance with class 2 of MIL-F-13927. After the test, the pump shall be capable of passing the test specified in 4.8.3.2 with no degradation.

4.8.3.4.3 Shock resistance: To determine conformance to 3.4.4.4, the pump shall be tested in accordance with method 213, test condition K (30 g amplitude and 11 ms time duration) of MIL-STD-202, except that only one shock in each direction of the three mutually perpendicular axes (total of six shocks) shall be applied. After the test, the pump shall be capable of passing the test specified in 4.8.3.2 with no degradation.

4.8.3.4.4 Vibration resistance. To determine conformance to 3.4.4.5, the pump shall be mounted as in intended operation and subjected to a simple harmonic motion having an amplitude of 0.8 mm (not more than 1.6 mm of total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 Hz. The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. The motion shall be applied for a period of 2 hr in each of 3 mutually perpendicular directions (total of 6 hr). After the test, the pump shall be capable of passing the test specified in 4.8.3.2 with no degradation.

4.8.3.4.5 Electromagnetic interference. To determine conformance to 3.4.4.5, the pump shall be tested for electromagnetic interference in accordance with MIL-STD-462, Notice 3.

4.8.3.5 Endurance. To determine conformance to 3.4.5, the pump shall pump water containing sodium chloride, sand and clay mixed in the following proportions and under the following conditions:

a. Mix 95 parts (by volume) fresh water and 5 parts sodium chloride.
b. For each 189 liters (L) of solution in a, add 2.3 kg of clay (Bentonite) and 0.9 kg of sand. The sand shall be 0.01 to 1.0 mm in diameter with the majority of grains 0.15 to 0.30 mm in diameter.
c. Circulation and agitation shall be such that the sand and clay remain in suspension throughout the test.
d. There shall be no maintenance or adjustment to be made to the pump or the motor during the test.
e. Test conditions shall be as specified in 4.8.3.2.

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4.8.3.5.1 Procedure. The endurance test shall be performed in the specified position (vertical or horizontal) as follows:

a. Water, as specified in 4.8.3.5, shall be pumped for 8 hr at 1.8 m water head and a flow of not less than 174 L/m.
b. The pump shall rest for 16 h.
c. Twelve cycles shall be performed as follows:

1. Three hours of pumping, as specified in step a.
2. Fifteen minute of dry-run.
3. A rest for 30 minutes.
4. Three hours of pumping, as specified in step a.
5. A rest for 16 1/2 h.

d. The above steps (a, c1 and c4) shall be performed 25 times for a total of 350 h of operating time. After the test, the rated capacity shall be determined in accordance with 4.8.3.2. The rated capacity shall be not less than 90% of that specified in 3.4.2.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump covered by this specification is intended to expel water and contaminations (sea water, fuels and lubricants) from the hulls of military, tactical and combat vehicles. The pump is used in the M113 personnel carrier.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable part number (see 1.2, 3.1, and 3.3).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. If first article inspection is not required (see 3.1).
e. If electric motor shall be other than as specified (see 3.3.5).
f. If finish should be other than specified (see 3.5).
g. If identification marking shall be other than as specified (see 3.6).

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h. If responsibility for inspections shall be other than as specified (see 4.1).
i. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
j. Type of first article sample and the number of samples to be inspected (see 4.4 and 6.3).
k. Arrangements for first article inspection, approval of test results and disposition of first article (see 6.3).
l. Government's rights and conditions for waiving first article inspection (see 6.3).
m. If lot formation shall be other than as specified (see 4.5.1.1).
n. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers on: whether the sample(s) should be a preproduction sample, a first article sample, an initial production sample, a first production item or a standard production item from the contractor's current inventory; the number of samples to be inspected as specified in 4.4; and (when applicable) the specific tests to be performed on each sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Static pressure head. Static pressure head is defined as the reading (converted to meters of water and corrected to the pump motor centerline) of a meter connected to a pressure tap in the wall of the discharge line, perpendicular to the direction of fluid flow, at a distance between 2.5 to 5 pipe diameters from the pump discharge outlet. The corrected meter reading shall include the pump inlet suction effects if the static suction head of the lift exceeds a distance of 10 cm below or above the pump centerline.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.3).

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6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Cross-reference. Pumps classified as type 1 in the previous revision are covered by this specification. Pumps classified as type 2 in the previous revision were deleted from this specification. However, type 2 pumps will be covered in a new specification, MIL-P-62415(AT).

6.6 Metrication. All units were converted to metric in accordance with FED-STD-376, revision A, dated 5 May 1983.

6.7 Subject term (key word) listing.

Axial flow water pump.
Bilge water pump.
Electric-motor-driven water pump.
Submersible water pump.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 4320-A278)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-62156B(AT); Pump, Submersible, Bilge, Axial Flow, Electric-Motor-Driven

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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